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Information Systems
Labwork 1

Section 1: Installing MySQL

I use CachyOS, a Linux distribution based on Arch Linux, so MySQL is not available in the default package manager. I can install it through the AUR (Arch User Repository) but I opted to use MariaDB, an open source alternative (forked from MySQL).

On Arch, MariaDB is currently at version 13.x, which is newer than the version that the test database is validated against (up to 12.x). Therefore, I chose the lts (long-term support) version which is currently at 12.x. I installed it through `shelly` (a wrapper for `pacman`). And there are some additional setup that is required to setup the database on the system (MariaDB Installation on Arch wiki).

```
$ shelly install standard mariadb-lts

$ sudo mariadb-install-db --user=mysql --basedir=/usr --datadir=/var/lib/mysql

Two all-privilege accounts were created.
One is root@localhost, it has no password, but you need to
be system 'root' user to connect. Use, for example, sudo mariadb
The second is mysql@localhost, it has no password either, but
you need to be the system 'mysql' user to connect.
After connecting you can set the password, if you would need to be
able to connect as any of these users with a password and without sudo

See the MariaDB Documentation at https://mariadb.com/docs

Please report any problems at https://mariadb.org/jira

The latest information about MariaDB is available at https://mariadb.org/.

Consider joining MariaDB's strong and vibrant community:
https://mariadb.org/get-involved/
```

Then I can start the service and check if the server is up and listening on `0.0.0.0`.

```
$ systemctl start mariadb.service
# to enable autostart: systemctl enable --now mariadb.service

mariadb.service - MariaDB 12.3.3 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/mariadb.service.d
             - galera.conf
     Active: active (running) since Tue 2026-11-03 19:28:18 +07; 20min ago
 Invocation: bc818816bfc24687a87d910b6eb02a78
       Docs: man:mariadbd(8)
             https://mariadb.com/docs/server/server-management/starting-and-stopping-mariadb/
                 systemd
   Main PID: 677018 (mariadbd)
     Status: "Taking your SQL requests now..."
```

```
      Tasks: 13 (limit: 312392)
     Memory: 283.9M (peak: 448.9M)
        CPU: 34.569s
     CGroup: /system.slice/mariadb.service
             - 677018 /usr/bin/mariadbd

Nov 03 19:28:17 ga401qh mariadbd[677018]: 2026-11-03 19:28:17 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/mysql/ib_buffer_pool
Nov 03 19:28:17 ga401qh mariadbd[677018]: 2026-11-03 19:28:17 0 [Note] InnoDB: Buffer pool(s) load completed at 261103 19:28:17
Nov 03 19:28:18 ga401qh mariadbd[677018]: 2026-11-03 19:28:18 0 [Note] Server socket created on IP: '0.0.0.0', port: '3306'.
Nov 03 19:28:18 ga401qh mariadbd[677018]: 2026-11-03 19:28:18 0 [Note] Server socket created on IP: '::', port: '3306'.
Nov 03 19:28:18 ga401qh mariadbd[677018]: 2026-11-03 19:28:18 0 [Note] mariadbd: Event Scheduler : Loaded 0 events
Nov 03 19:28:18 ga401qh mariadbd[677018]: 2026-11-03 19:28:18 0 [Note] /usr/bin/mariadbd: ready for connections.
Nov 03 19:28:18 ga401qh mariadbd[677018]: Version: '12.3.3-MariaDB'  socket: '/run/mysqld/mysqld.sock'  port: 3306  Arch Linux
Nov 03 19:28:18 ga401qh systemd[1]: Started MariaDB 12.3.3 database server.
```

Section 2: Set up account

With the database up and running, I need to set up access on local machine next. The first run required `sudo mariadb` because only `root` and `mysql` account have access to the database.

I created a user with the same name as my local account so I do not have to type `sudo` every time when I access the database from my local machine.

```
CREATE USER 'tom-maverick'@'localhost' IDENTIFIED VIA unix_socket;
GRANT ALL PRIVILEGES ON *.* TO 'tom-maverick'@'localhost';
FLUSH PRIVILEGES;
```

I also created another account for `tabularis`, an open source GUI database client that I use.

```
CREATE USER 'tabularis'@'localhost' IDENTIFIED BY 'tabularis';
GRANT ALL PRIVILEGES ON *.* TO 'tabularis'@'localhost';
FLUSH PRIVILEGES;
```

And we can check whether our account is set up correctly:

```
$ mariadb
MariaDB [(none)]> SELECT User, Host, plugin, authentication_string FROM mysql.user;
+--------------+-----------+-----------------------+-------------------------+
| User         | Host      | plugin                | authentication_string   |
+--------------+-----------+-----------------------+-------------------------+
| mariadb.sys  | localhost | mysql_native_password |                         |
| root         | localhost | mysql_native_password | invalid                 |
| mysql        | localhost | mysql_native_password | invalid                 |
| PUBLIC       |           |                       |                         |
|              | localhost |                       |                         |
|              | ga401qh   |                       |                         |
| tom-maverick | localhost | unix_socket           |                         |
| tabularis    | localhost | mysql_native_password | *231C7F49C35D199B57781B |
+--------------+-----------+-----------------------+-------------------------+
8 rows in set (0.001 sec)
```

Section 3: Import test data into MariaDB

With the set up finished, I can init the database with `test_db` data and verify the correctness by running the SHA2 test.

```
$ mariadb < test_employees_sha2.sql
INFO
TESTING INSTALLATION
table_name  expected_records  expected_crc
departments 9 377c5d727383a32633e2973f8e3411beffe29e2f4cc297c586fa6b24aa7df9ba
dept_emp  331603  34548ee9989dd4d5e065168b43249c8d3eb48bfbbfb3f2fc1cf01be6658f6a75
dept_manager  24  3a4e69723deec413a7d8a4f5ce55013830303fa617b6380ed2b0fd2d48b1c768
employees 300024  21f5d003842f24853e251d3d5116798bafe257ec3d1bb448b5365b68deaabbf4
salaries  2844047 4e99e691a9ea98fefc0b4fec8ca4e758baeefba2967bd8d6474810a9a5f6e729
titles  443308  a9e940ef9ba1029a8f0356fdbe495430bedc59eec5ceb4f71e0cc35ddcbf9980
table_name  found_records found_crc
departments 9 377c5d727383a32633e2973f8e3411beffe29e2f4cc297c586fa6b24aa7df9ba
dept_emp  331603  34548ee9989dd4d5e065168b43249c8d3eb48bfbbfb3f2fc1cf01be6658f6a75
dept_manager  24  3a4e69723deec413a7d8a4f5ce55013830303fa617b6380ed2b0fd2d48b1c768
employees 300024  21f5d003842f24853e251d3d5116798bafe257ec3d1bb448b5365b68deaabbf4
salaries  2844047 4e99e691a9ea98fefc0b4fec8ca4e758baeefba2967bd8d6474810a9a5f6e729
titles  443308  a9e940ef9ba1029a8f0356fdbe495430bedc59eec5ceb4f71e0cc35ddcbf9980
table_name  records_match crc_match
departments OK  ok
dept_emp  OK  ok
dept_manager  OK  ok
employees OK  ok
salaries  OK  ok
titles  OK  ok
summary result
CRC OK
count OK
```

And then we can use `mariadb` client to check the table within the `employees` database.

```
$ mariadb
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 12
Server version: 12.3.3-MariaDB Arch Linux

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE employees;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [employees]> SHOW TABLES;
+----------------------+
| Tables_in_employees  |
+----------------------+
| current_dept_emp     |
| departments          |
| dept_emp             |
| dept_emp_latest_date |
| dept_manager         |
| employees            |
| salaries             |
| titles               |
+----------------------+
8 rows in set (0.001 sec)
```

```
MariaDB [employees]> SELECT * from departments;
+---------+--------------------+
| dept_no | dept_name          |
+---------+--------------------+
| d009    | Customer Service   |
| d005    | Development        |
| d002    | Finance            |
| d003    | Human Resources    |
| d001    | Marketing          |
| d004    | Production         |
| d006    | Quality Management |
| d008    | Research           |
| d007    | Sales              |
+---------+--------------------+
9 rows in set (0.001 sec)
```